API architecture review checklist

API Platform

6 November, 2026

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To use this page:

- Create a copy of it in an appropriate place in your own Confluence space
- Rename it to "<My API name> API architecture review"
- Add a link to it in API register[1]
- Fill out the page - deleting anything you don't need, such as this tip. When you edit this page there are number of hints and further help.

This template page is to:

- give advice on how to architect an API on the NHS Digital API Platform
- provide a structure for basic meta data about your API that the Platform team will use to populate / manage the platform
- challenge you early on about the consistency of your API with other NHS Digital APIs

CONSIDER time-boxing filling in this page to a max of TWO hours prior to an API architecture review

1. https://nhsd-confluence.digital.nhs.uk/display/APM/API+register

1 Overview

This page is the API architecture review checklist for the above API.

It captures (or provides links to) details of key architectural decisions for the API.

Links might be to documentation on other Confluence pages, or possibly code artefacts, such as the API's OAS file.

As per the API delivery process[2], the decisions are split into two sections:

1. in the discovery phase: **architecture** - big architectural decisions
2. in the alpha phase: **design** - detailed design decisions

2. https://nhsd-confluence.digital.nhs.uk/display/APM/Deliver+your+API

2 Status codes key

TO DO

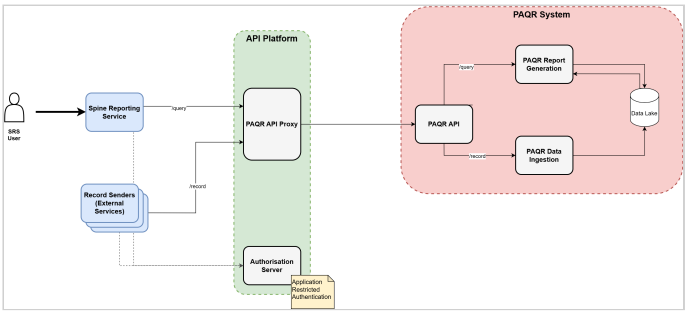
IN PROGRESS

DONE

N/A

3 Architecture checklist (discovery phase)

If this API is going to TRG for review, it might make sense to complete an Solution Design Options (SDO) and / or Key Architectural Decision (KAD) document and link out to that.

Activity	Status	Comments / Evidence
Explain the context / problem / opportunity space for your API.	TO DO	Sharing Clinical Data between different organisations.
Consider solution / architecture options to address the problem / opportunity space, and select a preferred option. Everything below this activity is based on the selected option, and assumes the selected option includes an API.	TO DO	<provide evidence of your options, for example by completing a copy of our API architecture options template[3] and linking out to that>
Identify the key use cases / functions for your API.	TO DO	<complete as appropriate or link out to details elsewhere> This API will allow external systems to: • TBC function 1 • TBC function 2 • etc
Identify the end user - healthcare worker, patient or not present.	TO DO	<complete as appropriate or link out to details elsewhere> The end user is a **healthcare worker** / **patient** / **not present**
Determine the end-to-end architecture - how data flows and where processing occurs. Illustrate it using a context diagram.	TO DO	<edit the diagram below or link out to details elsewhere>  <if necessary, add a narrative>

3. https://nhsd-confluence.digital.nhs.uk/display/APM/API+architecture+options+template

Activity	Status	Comments / Evidence
If your API routes requests to multiple backends, figure out how it will do this. • See Routing requests to multiple backends[4].	TO DO	<incorporate your decisions into the above diagram> See above diagram.
Determine where the various components of your architecture (including your back-end, if new) will be hosted. • We have some hosting patterns - we can help you with this. https://nhsd-confluence.digital.nhs.uk/display/APM/Transforming+data+on+the+APIM+platform+and+containers[5]	TO DO	<provide details, or draw a diagram, or link out to details elsewhere>
Decide on the API consumer security pattern. • See API consumer security[6]	TO DO	<edit as appropriate or link out to details elsewhere> This API uses the following security pattern(s): <delete as applicable> • application-restricted access - using API key authentication (as per API consumer security - Application restricted- API key authentication[7]) • application-restricted access - using signed JWT authentication (as per API consumer security - Application restricted API- signed JWT authentication[8]) • user-restricted access - using NHS Identity for end user authentication (end user is a healthcare professional) (as per API consumer security- User-restricted-NHS CIS2[9]) • user-restricted access - using NHS Login for end user authentication (end user is a citizen) (as per API consumer security- User-restricted-NHS login[10]) This pattern is appropriate because: • <justify the decision>

4. https://nhsd-confluence.digital.nhs.uk/display/APM/Routing+requests+to+multiple+backends
5. https://gbr01.safelinks.protection.outlook.com/?url=https%3A%2F%2Fnhsd-confluence.digital.nhs.uk%2Fdisplay%2FAPM%2FTransforming%2Bdata%2Bon%2Bthe%2BAPIM%2Bplatform%2Band%2Bcontainers&data=05%7C01%7Cpaul.fisher18%40nhs.net%7C576a3f84242849e79d3708db14cfca35%7C37c354b285b047f5b22207b48d774ee3%7C0%7C0%7C638126654897068216%7CUnknown%7CTWFpbGZsb3d8eyJWIjoiMC4wLjAwMDAiLCJQIjoiV2luMzIiLCJBTiI6Ik1haWwiLCJXVCI6Mn0%3D%7C3000%7C%7C%7C&sdata=fmfLXuxADYr8tJHeq39tTxppBJAf2B2Ltq8Jpl5omeM%3D&reserved=0
6. https://nhsd-confluence.digital.nhs.uk/display/APM/API+consumer+security
7. https://nhsd-confluence.digital.nhs.uk/display/APM/API+consumer+security#APIconsumersecurity-Application-restrictedAPI-APIkeyauthentication

Activity	Status	Comments / Evidence
Decide how to do backend security. Security for API Backend[11]	TO DO	<edit as appropriate or link out to details elsewhere> The API proxy connection to the backend is secured using **TLS-MA** / **<something else>**.
Decide on the integration pattern for the API. • See Integration Pattern Book[12] • Your APIs should be synchronous under the majority of situations	TO DO	<edit as appropriate or link out to details elsewhere> This API uses the following integration pattern(s): <delete, amend or add as appropriate> • REST - synchronous <**recommend**> • REST - asynchronous short polling • some form of messaging pattern
Decide whether to use FHIR. • See Is my API a FHIR API?[13]. • See API Landscape[14].	TO DO	<edit as appropriate or link out to details elsewhere> This API is / is not FHIR-conformant. In particular: • It uses **FHIR R4** • It uses the following FHIR R4 global base resources: **TBC** • It has the following local profiles in FHIR UK Core: **TBC**
If you're not using FHIR but you should be, or if you're using an older version fo FHIR, explain why. You'd better have a damn good reason.	TO DO	<complete as appropriate or link out to details elsewhere>
Decide on the payload type(s). • See https://developer.nhs.uk/apis/fhir-policy/serialisation.html. • Consider large payloads - see Large payloads[15].	TO DO	<edit as appropriate or link out to details elsewhere> The API payload is **JSON** / **XML**. Will any payload exceed 10MB in size? **YES** / **NO**

8. https://nhsd-confluence.digital.nhs.uk/display/APM/API+consumer+security#APIconsumersecurity-Application-restrictedAPI-signedJWTauthentication
9. https://nhsd-confluence.digital.nhs.uk/display/APM/API+consumer+security#APIconsumersecurity-User-restricted-NHSCIS2
10. https://nhsd-confluence.digital.nhs.uk/display/APM/API+consumer+security#APIconsumersecurity-User-restricted-NHSlogin
11. https://nhsd-confluence.digital.nhs.uk/display/APM/Security+for+API+Backend
12. https://digital.nhs.uk/about-nhs-digital/our-work/nhs-digital-architecture/integration-patterns-book
13. https://nhsd-confluence.digital.nhs.uk/pages/viewpage.action?pageId=216171050
14. https://nhsd-confluence.digital.nhs.uk/display/APM/API+Landscape
15. https://nhsd-confluence.digital.nhs.uk/display/APM/Large+payloads

<table>
<tr><th>Activity</th><th>Status</th><th>Comments / Evidence</th></tr>
<tr><td>Decide the API's service level. These specify SLAs around things like availability and incident fix times.
See Service levels for APIs[16].</td><td>TO DO</td><td><edit as appropriate or link out to details elsewhere>
During private beta, the service level for this API is BRONZE.
From public beta and beyond, the service level for this API is PLATINUM.</td></tr>
<tr><td>Determine what path-to-live environments you'll need to meet your API consumers' testing needs.</td><td>TO DO</td><td><edit as appropriate or link out to details elsewhere>
The API includes the following path-to-live environments:
<table>
<tr><th>Environment</th><th>Purpose</th><th>API consumer security pattern(s)</th><th>Implementation notes</th></tr>
<tr><td>Integration test</td><td>Developer initial work
Formal integration testing</td><td>As per production</td><td>TBC which backend environment does this connect to?
TBC how will test data be managed?</td></tr>
<tr><td>Sandbox</td><td>Early developer investigation</td><td>Open access</td><td>Canned responses to a small number of scenarios.
Response fragments shared with API specification.
Deployed to AWS as per API platform standard approach.
Stateless.</td></tr>
</table>
</td></tr>
</table>

16. https://nhsd-confluence.digital.nhs.uk/display/APM/Service+levels+for+APIs

<table>
<tr><th>Activity</th><th>Status</th><th>Comments / Evidence</th></tr>
<tr><td>Decide the granularity and name(s) for your API(s).
See API naming and grouping endpoints[17]</td><td>TO DO</td><td><edit as appropriate or link out to details elsewhere>
The functionality described on this page is grouped into coherent APIs as follows:
<table><tr><th>Plain English name (as per the API catalogue[18])</th><th>Physical name (URL)</th><th>Functional scope</th></tr><tr><td>Do A Thing Service - FHIR API</td><td>https://api.service.nhs.uk/do-a-thing/FHIR/R4</td><td>All functions</td></tr></table></td></tr>
<tr><td>Check that your API doesn't overlap with another API owned by another team.
The above naming exercise, and a scan of the API catalogue[19], will help with this.
Silo teams building overlapping APIs has historically been one of our biggest issues.</td><td>TO DO</td><td><confirm or discuss as appropriate></td></tr>
</table>

17. https://nhsd-confluence.digital.nhs.uk/display/APM/API+naming+and+grouping+endpoints
18. https://digital.nhs.uk/developer/api-catalogue
19. https://digital.nhs.uk/developer/api-catalogue

<table>
<tr><th>Activity</th><th>Status</th><th>Comments / Evidence</th></tr>
<tr><td>Estimate the expected volumes for your API.</td><td>TO DO</td><td><edit as appropriate or link out to details elsewhere>
We estimate that the API will experience the following volumes:
<table><tr><th>Metric</th><th>Private beta</th><th>Long term</th></tr><tr><td>Number of connected systems</td><td>TBC</td><td>TBC</td></tr><tr><td>Peak load transactions per second (TPS) per connected system</td><td></td><td></td></tr><tr><td>Peak load transactions per second (TPS) overall</td><td></td><td></td></tr><tr><td>Transactions per month per connected system</td><td></td><td></td></tr><tr><td>Transactions per month overall</td><td></td><td></td></tr></table></td></tr>
<tr><td>Do you have PID or sensitive data in your URL Path or Query parameters?
As key pieces of data, such as NHS number, is a key element of any RESTful design, the platform is geared up to have sensitive data in this.
However, this should be kept to a minimum because of systems (e.g. client browsers) that have access to the full values could log such data</td><td>TO DO</td><td><discuss as appropriate - there are a number of pages in the Producer Zone, for instance Privacy considerations in your design[20]></td></tr>
</table>

20. https://nhsd-confluence.digital.nhs.uk/display/APM/Privacy+considerations+in+your+design

4 Design checklist (alpha phase)

4.1 Consistency checklist

The API Management Platform is going to be a key entry point into NHS Digital, and as a consequence will highlight potential inconsistency between different approaches. Some of these consistencies can be hidden behind the Platform façade, others simply won't be cost effective to change. See the page on Consistency[21] for a run down of the wider view of this issue.

The following checklist is designed to help you identify inconsistencies early on.

4.1.1 Primary considerations

For detailed guidance on what is meant in these areas see: Designing your API[22]

Area	Consideration		If No:
API Name	Does it fit into the wider APIM Namespace / domain model?	Y/N	*For guidance on naming see:* API naming and grouping endpoints[23]
Session	Can Apigee manage the session with your API Consumer? Does the API Consumer call other APIs outside of Apigee?	Y/N	This has a number of small complications, so if your existing API has a custom call to start the session - or lots of data is pre-configured during authentication or authorisation this might need some extra design work
Update	PUT is for updating entire FHIR resources PATCH is being used where there are small changes to a large FHIR resource	Y/N	
Atomic operations	Always use small atomic (stateless) requests - batching multiple items should be a last resort. Batching encourages poor design, reduces scalability and increases the 'blast radius' of any failures	Y/N	• Are the requests related by a transaction? This could (would) then still be atomic - but it depends on the size of the batch. • Is there a cost to enabling atomic over batch?

21. https://nhsd-confluence.digital.nhs.uk/display/APM/Consistency
22. https://nhsd-confluence.digital.nhs.uk/display/APM/Designing+your+API
23. https://nhsd-confluence.digital.nhs.uk/display/APM/API+naming+and+grouping+endpoints

Area	Consideration		If No:
Data minimisation	Do your endpoints return the minimum amount of data possible? (To align with GDPR and other design principles)	Y/N	If not, could you: • Default to a smaller subset, and allow parameter to expand for specific requests? • Could you have different access modes, so the lower security connections receive less data
Testing	Are your current unit and integration tests all automated?	Y/N	
	Can you perform the testing via Apigee and your proxy?	Y/N	This is important to ensure the end to end service is fully tested.

4.1.2 Secondary considerations

These are the next level down of detail, and ideal for exploring in Alpha and Private Beta.

Area	Consideration		Response
Terminology	Are there key terms in your API that don't fit with other NHS Digital services?		
URL Path	Are there terms that might be confusing, stand out from other APIs? Is the path very long? See policy APIM-LAND-05 on API Landscape[24] for details of requirements for FHIR API URL Paths.		
Endpoint names	Use nouns instead of verbs (RESTful pattern) Are they intuitive? For FHIR APIs, endpoint names must align with the FHIR resource model as described at https://www.hl7.org/fhir/http.htmland https://www.hl7.org/fhir/resourcelist.html (the endpoint name must be capitalised) For non-FHIR APIs use spinal-case		*If there are inconsistencies here, mapping of these in Apigee is easy. However, what impacts are there on your existing users?*

24. https://nhsd-confluence.digital.nhs.uk/display/APM/API+Landscape

Area	Consideration		Response
Query Parameters	Are these consistent with other API Platform query parameters? FHIR APIs must be conformant with https://www.hl7.org/fhir/search.html		*If there are inconsistencies here, mapping of these in Apigee is easy. However, what impacts are there on your existing users?*
Field Names	Are the field names in the **payload** datasets the same as FHIR?		*Even for non-FHIR APIs, should developers have to deal with a different field name for basic data such as first name and last name?*
REST Conventions	Does your HTTP Verb usage align with RESTful principles? For FHIR APIs, does this also align with https://www.hl7.org/fhir/http.html		
	Do your HTTP Status Codes align with RESTful principles?		
	Do your FHIR APIs align with guidance found in the FHIR specification for HTTP status codes		*tbc guidance on this side*
Headers	For API Consumers - are they consistent with other API Management Headers?		*You decide as an API Publisher which platform headers are mandatory, see* HTTP headers for your API[25] *for detailed guidance*
	For your API backend (less important as you are managing this)		
Environments	Does the way you use environments align with the API Management platform, and the Path To Live?		See https://confluence.digital.nhs.uk/display/APM/Architecture#Architecture-environments
API Versioning	The private beta hasn't needed to deal with versioning, but we are looking to use accept-headers to support this in the future, though the current NHS Digital policies include the API major version in the URL path.		

25. https://nhsd-confluence.digital.nhs.uk/display/APM/HTTP+headers+for+your+API

Area	Consideration		Response
Error Messages	How you return error messages should be consistent with other interfaces (e.g. FHIR APIs will return OperationOutcome resources to express error detail)		See additional information pages here Handling errors[26] Consistency between APIs[27] RESTful HTTP design points[28]
Test Data	What is your approach to test data for External Developers?	-	
S Flag	Does your API need to consider the Sensitive flag status[29]	Y/N	

4.1.3 Detailed design considerations

The following are very detailed elements about your APIs, but as with the other points above, they can introduce breaking changes into your design. The design pattern in place on the Platform has come out of a combination between the FHIR standard, Apigee platform design, Spine as a key backend across the platform, and input from across the Tribe and NHS Digital.

Area	Consideration		
ETag	FHIR standard mandates the use of the ETag for version checking		See Does my API need to use ETag Headers[30]
Postman collections	There are other Postman collections across the platform, does yours follow the same conventions ?		
Datetime	Date and time field values should follow the ISO-8601[31] standard, read the details on how to do this: RESTful HTTP design points[32]		

26. https://nhsd-confluence.digital.nhs.uk/display/APM/Handling+errors
27. https://nhsd-confluence.digital.nhs.uk/display/APM/Consistency+between+APIs
28. https://nhsd-confluence.digital.nhs.uk/display/APM/RESTful+HTTP+design+points
29. https://digital.nhs.uk/services/national-back-office-for-the-personal-demographics-service/management-of-nhs-numbers-and-pds-records
30. https://nhsd-confluence.digital.nhs.uk/display/APM/Update+locking+with+ETag
31. https://en.wikipedia.org/wiki/ISO_8601
32. https://nhsd-confluence.digital.nhs.uk/display/APM/RESTful+HTTP+design+points

5 Governance

This section includes information related to the architecture and design reviews that occur as part of the API delivery process[33].

5.1 API architecture review

Date	TO DO
Attendees	TO DO
Actions	TO DO

5.2 API design review

Update this page with the latest, could be important points for the later TRG / LSB governance steps

Date	TO DO
Attendees	TO DO
Actions	TO DO

5.3 Post implementation review

Update with any final changes, and flag up any key future work

Date	TO DO
Attendees	TO DO
Actions	TO DO

33. https://nhsd-confluence.digital.nhs.uk/display/APM/Deliver+your+API